

When Do Class-Imbalance Corrections Succeed? A Controlled Comparison of Weighting, Resampling, and Threshold Selection Across Disease Prevalence Levels

Adam Issac

Denmark High School, Alpharetta, Georgia, USA

Correspondence: aissac8@gatech.edu

Abstract

Class imbalance is one of the most common obstacles in clinical machine learning, and several corrective techniques are in routine use, yet practitioners receive little systematic guidance on which correction to use with which model, or at what degree of imbalance corrections matter at all. This study characterized when five standard strategies succeed or fail. Datasets with controlled disease prevalence of 2, 5, 10, 20, 35, and 50 percent ($N = 20,000$ each) were constructed by stratified subsampling from the full CDC Behavioral Risk Factor Surveillance System 2020 heart disease dataset (319,795 records), with ten random replicates per level. Three model families (Logistic Regression, Random Forest, and Gradient Boosting) were crossed with six strategies: default training, balanced class weighting, random undersampling, SMOTE, decision-threshold selection on a held-out validation split, and weighting combined with threshold selection, for 1,080 evaluations. Default models failed at low prevalence in every family (balanced accuracy 0.50 at 2 percent prevalence) and converged with corrected models at 50 percent. Corrections were strongly model dependent: class weighting failed completely for Random Forest at every prevalence and was significantly worse than no correction at 10 percent ($p = 0.014$), and SMOTE failed for both tree ensembles while matching weighting for Logistic Regression. Validation-tuned thresholding was the only strategy effective for every model at every prevalence, at zero retraining cost. ROC-AUC was nearly invariant across training strategies, indicating that corrections alter the operating point rather than the learned ranking, and the selected threshold approximately equaled the prevalence for default-trained models. Two additional experiments confirmed and extended these results. Replication on two further BRFSS-independent datasets (a continuous-feature cardiovascular dataset and a binary-feature diabetes dataset) reproduced every headline finding and confirmed a predicted mechanism: SMOTE's failure for tree ensembles is specific to predominantly binary feature spaces and is substantially relieved when continuous features dominate. Adding focal loss, an asymmetric training objective, placed it on par with class weighting but still below threshold tuning. These results yield practical rules: correct only below roughly 20 percent prevalence, avoid class weighting and, for binary feature spaces, SMOTE with tree ensembles, and treat class imbalance primarily as an operating-point problem.

Keywords: class imbalance, prevalence, decision threshold, class weighting, SMOTE, undersampling, heart disease, BRFSS

1. Introduction

Cardiovascular diseases are the leading cause of death globally, accounting for an estimated 17.9 million deaths each year [1], and machine learning models trained on routinely collected health data are increasingly proposed as screening aids [2]. Large public surveys such as the Behavioral Risk Factor Surveillance System (BRFSS) make population-scale behavioral and demographic

data widely available for such models [3]. These data share a structural property with most clinical prediction problems: the condition of interest is rare, so the classes are heavily imbalanced.

The consequences of imbalance for naively trained classifiers are well documented. Models trained with default settings and a 0.5 decision threshold collapse toward the majority class, producing high accuracy and near-zero sensitivity [4], and accuracy-style metrics conceal the failure [5]. A substantial literature offers remedies: cost-sensitive learning and class weighting, synthetic minority oversampling (SMOTE) [6], majority undersampling, and decision-theoretic threshold adjustment [7]. Recent work has also warned that imbalance corrections are not free, showing that they can distort the calibration of clinical risk models [8].

What this literature offers less of is guidance on the conditions under which each remedy works: which corrections are effective for which model families, how effectiveness depends on the degree of imbalance, and at what prevalence corrections stop mattering altogether. Most empirical comparisons evaluate a single dataset at its native prevalence, which confounds the correction, the model, and the imbalance level.

This study addresses that gap with a controlled factorial design. Disease prevalence is manipulated directly, from 2 to 50 percent, by stratified subsampling from a single large clinical survey dataset, holding the feature space and sample size fixed. Three model families are crossed with six imbalance strategies at each prevalence level, with ten random replicates, and decision thresholds are always selected on a held-out validation split rather than the test set. The goals are to (1) map when default training fails as a function of prevalence, (2) identify which corrections succeed or fail for which model family, and to test whether those differences are statistically reliable, and (3) derive general, practically usable principles about class-imbalance correction, including the mechanism behind the observed failures.

2. Related Work

He and Garcia survey the main families of imbalance remedies: resampling, cost-sensitive methods, and ensemble variants [4]. Chawla et al. introduced SMOTE, which synthesizes minority examples by interpolating between minority-class neighbors [6]. Elkan established the decision-theoretic view in which class priors and misclassification costs determine an optimal decision threshold, implying that many imbalance problems can be addressed at the operating point rather than during training [7]. For Random Forests specifically, Chen, Liaw, and Breiman found that weighting-based corrections are less effective than per-tree balanced sampling [9]. More recently, Elor and Averbuch-Elor questioned whether SMOTE benefits strong classifiers at all [10], and van den Goorbergh et al. showed that imbalance corrections can harm the calibration of logistic regression risk models, recommending against routine correction [8]. On evaluation, Saito and Rehmsmeier demonstrate that precision-recall analysis is more informative than ROC analysis under severe skew [5]. The present study connects these threads in a single controlled design: it tests weighting, two resampling methods, and validation-based threshold selection side by side, across model families and a full range of prevalence levels, on one large clinical dataset.

3. Materials and Methods

3.1 Dataset and Controlled Prevalence Construction

The primary experiments used the full Personal Key Indicators of Heart Disease dataset [11], derived from the CDC BRFSS 2020 survey [3]: 319,795 records, 17 predictors spanning health

behaviors, medical history, and demographics, and a binary self-reported heart disease target with a native prevalence of 8.56 percent (27,373 positive records). For each target prevalence p in $\{2, 5, 10, 20, 35, 50\}$ percent, datasets of fixed size $N = 20,000$ were constructed by sampling, without replacement, $\text{round}(pN)$ positive and $N - \text{round}(pN)$ negative records. Ten datasets per prevalence level were drawn with different random seeds, so that all reported results are means over ten independent replicates. Because the number of positives available (27,373) exceeds the maximum required (10,000 at 50 percent prevalence), every constructed dataset draws real records only.

To test whether the findings generalize beyond a single dataset and feature space, two additional datasets were used for replication (Section 4.5). The first is a cardiovascular disease dataset of 70,000 clinical examination records [14] whose feature space is continuous-dominant (five continuous measurements including blood pressure and body metrics, six categorical). The second is the BRFSS 2015 diabetes health-indicators dataset of 253,680 records [15], whose feature space is binary-dominant (seven continuous, fourteen binary), analogous in structure to the primary dataset but drawn from a different survey year and target condition. The identical controlled-prevalence protocol was applied to both at prevalences of 2, 10, and 35 percent with five seeds each. These two datasets bracket the primary dataset in continuous-feature share, which is the property the SMOTE mechanism proposed in Section 4.2 predicts should matter.

3.2 Preprocessing and Data Splits

Preprocessing was identical at every design point. The target was mapped from Yes/No to 1/0; the 13 categorical predictors were one-hot encoded with the first level dropped, yielding 37 features (4 continuous, 33 binary indicators); and each constructed dataset was split into stratified training (60 percent), validation (20 percent), and test (20 percent) partitions. Continuous standardization parameters were fit on the training partition only and applied to the validation and test partitions. All threshold selection used the validation partition exclusively; the test partition was touched only for final evaluation.

3.3 Models

Three model families were implemented with scikit-learn [12]: Logistic Regression (L2 regularization, lbfgs solver), Random Forest (300 trees), and Gradient Boosting (HistGradientBoostingClassifier with library defaults), each with fixed random seeds. These represent three distinct decision mechanisms: a linear loss-minimizing model, a bagged majority-vote ensemble, and a boosted loss-minimizing ensemble. A preliminary single-prevalence study by the author had suggested that Random Forest responds differently to class weighting than linear models do; the present design tests that observation systematically.

3.4 Imbalance Strategies

Six strategies were evaluated for every model at every prevalence (Table 1). Balanced class weighting uses scikit-learn's `class_weight` option, which weights each class inversely to its frequency. Random undersampling and SMOTE, both from imbalanced-learn [13], rebalance the training partition to a 1:1 ratio; both were applied to the training partition only, after standardization. The threshold strategy trains the default model and then selects the decision threshold that maximizes Youden's J statistic (sensitivity plus specificity minus 1) on the validation partition, and the combined strategy applies the same validation-based selection to the class-

weighted model. Selecting thresholds on validation data, rather than on the test set, ensures the reported operating points are honest out-of-sample estimates.

Because focal loss is a widely cited asymmetric alternative to class weighting [16], it was added as a further training-time strategy in a dedicated experiment (Section 4.6). Focal loss reshapes the cross-entropy objective by a factor $(1 - p_t)^\gamma$ raised to a focusing parameter γ , down-weighting well-classified examples so that training concentrates on hard, typically minority, cases. Two implementations were used, both with $\gamma = 2$ and an inverse-frequency class-balance term. For Logistic Regression, the focal objective was minimized directly with L-BFGS using an analytic gradient (verified against finite differences to a maximum error below $1e-5$). For a boosted-tree implementation, LightGBM [17] was trained with a custom focal objective supplying the gradient and a finite-difference Hessian, and compared against LightGBM with default, class-weighted, and validation-tuned-threshold configurations on the primary dataset across all six prevalence levels with ten seeds.

Table 1. The six strategies evaluated for every model family at every prevalence level.

Strategy	Description
Default	Model trained as-is; positive prediction when predicted probability ≥ 0.5 .
Class weight	Retrained with balanced class weights, penalizing errors on each class inversely to its frequency.
Undersample	Majority class randomly undersampled to a 1:1 ratio in the training partition before fitting.
SMOTE	Minority class synthetically oversampled to a 1:1 ratio in the training partition before fitting.
Tuned threshold	Default-trained model; decision threshold selected on the validation partition by maximizing Youden's J.
Weight + tuned threshold	Class-weighted model combined with a validation-selected Youden threshold.

3.5 Evaluation and Statistical Analysis

Each model-strategy combination was evaluated on the untouched test partition ($n = 4,000$) using sensitivity, specificity, balanced accuracy, positive-class precision and F1, ROC-AUC, and average precision (PR-AUC), computed from predicted probabilities where applicable. The full design comprises 6 prevalences $\times$ 10 seeds $\times$ 3 models $\times$ 6 strategies = 1,080 evaluations from 720 fitted models. Results are reported as means with standard deviations over the ten seeds. Pre-specified headline contrasts were tested with two-sided paired Wilcoxon signed-rank tests across the ten seeds; with $n = 10$, the smallest attainable p-value is 0.002. No multiple-comparison correction was applied, and p-values are reported descriptively for a small set of planned contrasts.

4. Results

4.1 Default Training Fails at Low Prevalence for Every Model Family

Figure 1 and Table 2 summarize balanced accuracy across the design. At 2 percent prevalence, all three default-trained models performed at chance: balanced accuracy 0.504 (Logistic Regression), 0.500 (Random Forest), and 0.502 (Gradient Boosting), with sensitivities of 0.9, 0.1, and 0.4 percent respectively. Default performance climbed steadily with prevalence and converged with

every corrected strategy at 50 percent (0.742 to 0.762 across models). The spread between the best and worst corrective strategies, 0.21 to 0.24 balanced-accuracy points at 2 percent prevalence for the tree ensembles, collapsed to 0.017 to 0.019 at 35 percent and to 0.001 at 50 percent. Class-imbalance correction is therefore essential below roughly 20 percent prevalence and immaterial at or above 35 percent on this data.

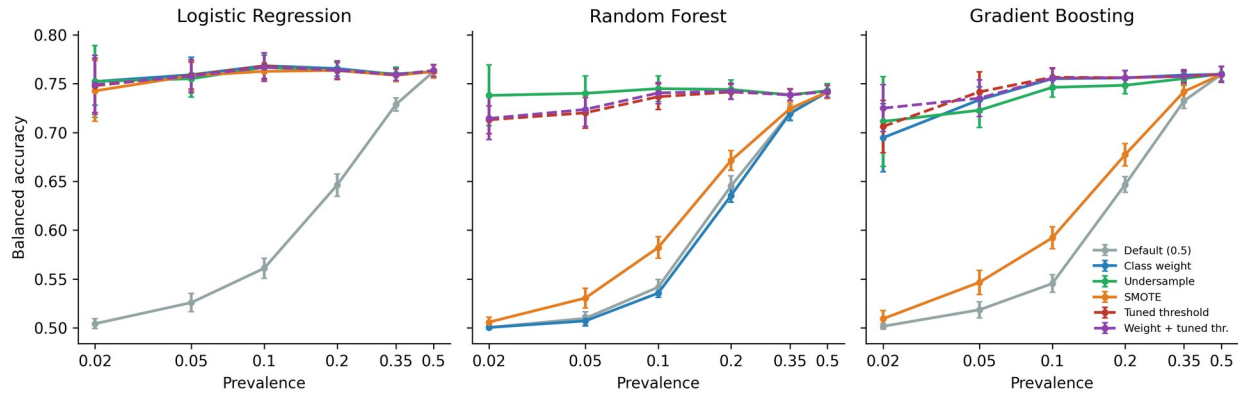


Figure 1. Balanced accuracy (mean and standard deviation over 10 seeds) as a function of prevalence, for each model family and strategy. Note the Random Forest panel: the class-weight line lies on the default line at every prevalence, while SMOTE tracks the default closely for both tree ensembles.

Table 2. Balanced accuracy, mean (standard deviation) over 10 seeds, at three representative prevalence levels.

Model	Strategy	Prevalence 2%	Prevalence 10%	Prevalence 35%
Logistic Regression	Default	0.504 (0.005)	0.561 (0.010)	0.729 (0.007)
	Class weight	0.752 (0.024)	0.768 (0.013)	0.760 (0.007)
	Undersample	0.752 (0.037)	0.767 (0.012)	0.760 (0.007)
	SMOTE	0.743 (0.031)	0.763 (0.009)	0.759 (0.007)
	Tuned threshold	0.748 (0.028)	0.768 (0.014)	0.759 (0.006)
	Weight + tuned thr.	0.749 (0.030)	0.767 (0.015)	0.759 (0.006)
Random Forest	Default	0.500 (0.002)	0.542 (0.008)	0.720 (0.008)
	Class weight	0.500 (0.002)	0.536 (0.004)	0.719 (0.007)
	Undersample	0.738 (0.031)	0.745 (0.013)	0.739 (0.006)
	SMOTE	0.506 (0.005)	0.582 (0.011)	0.724 (0.006)
	Tuned threshold	0.713 (0.014)	0.737 (0.013)	0.739 (0.006)
	Weight + tuned thr.	0.714 (0.022)	0.741 (0.011)	0.738 (0.006)
Gradient Boosting	Default	0.502 (0.003)	0.545 (0.009)	0.732 (0.008)
	Class weight	0.695 (0.035)	0.755 (0.011)	0.759 (0.005)
	Undersample	0.711 (0.046)	0.746 (0.010)	0.755 (0.007)
	SMOTE	0.509 (0.008)	0.592 (0.011)	0.742 (0.008)
	Tuned threshold	0.706 (0.027)	0.757 (0.009)	0.758 (0.005)
	Weight + tuned thr.	0.725 (0.024)	0.756 (0.011)	0.758 (0.007)

4.2 Which Corrections Work Depends on the Model Family

The corrections separated sharply by model (Table 2, Figure 2). For Logistic Regression, every correction worked at every prevalence: at 2 percent, balanced accuracy was 0.743 to 0.752 across all five corrective strategies, and SMOTE was statistically indistinguishable from class weighting (0.743 vs. 0.752, $p = 0.11$). For Random Forest, class weighting failed completely: balanced accuracy 0.500 at 2 percent, identical to default, and at 10 percent it was significantly worse than applying no correction at all (0.536 vs. 0.542, $p = 0.014$). SMOTE likewise failed for both tree ensembles at low prevalence (Random Forest 0.506, Gradient Boosting 0.509 at 2 percent), while random undersampling succeeded for the same models (0.738 and 0.711; both SMOTE-vs-undersampling contrasts $p = 0.002$). Gradient Boosting, in contrast to Random Forest, did respond to class weighting (0.695 at 2 percent, 0.755 at 10 percent). Sensitivity results mirror these patterns: at 2 percent prevalence, default models detected 0.1 to 0.9 percent of true cases, while effective corrections detected 54 to 78 percent, with all but one (Gradient Boosting class weighting, at 54 percent) falling between 72 and 78 percent.

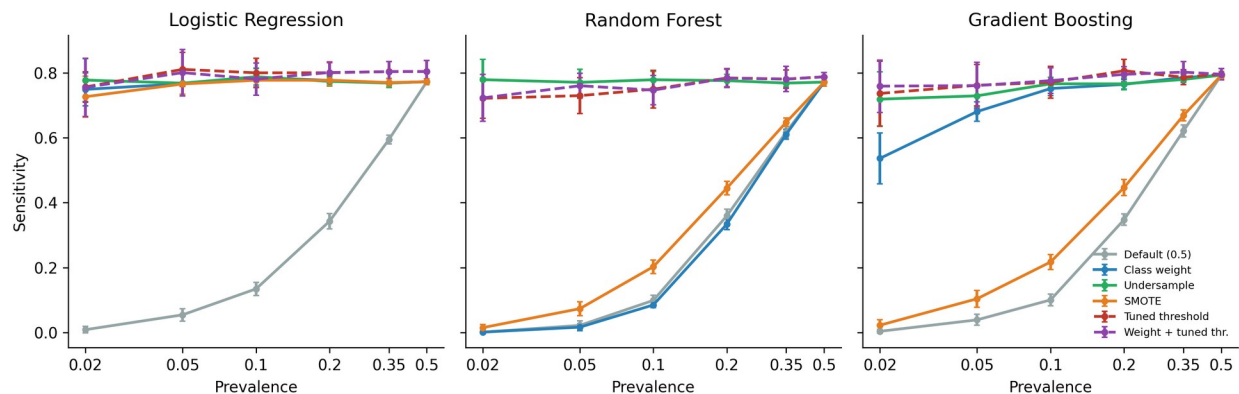


Figure 2. Sensitivity (mean and standard deviation over 10 seeds) as a function of prevalence, for each model family and strategy.

Two aspects of these failures are worth emphasizing. First, the Random Forest weighting failure is total and uniform: at no prevalence did balanced class weighting move the ensemble away from default behavior, and undersampling, which rebalances the data prior directly, fixed the same model at every prevalence, discarding up to 96 percent of the negative training records at 2 percent prevalence to do so. Second, the SMOTE failure is specific to the tree ensembles. A plausible mechanism follows from the feature space: after one-hot encoding, 33 of the 37 features are binary indicators, and SMOTE's linear interpolation between minority neighbors produces fractional values on those indicators that no real record can take. A linear decision function still shifts sensibly under such synthetic mass, but axis-aligned tree splits generalize poorly from impossible feature values, and the class prior seen by the trees barely changes in the regions that matter. This account is consistent with recent evidence that SMOTE provides little benefit for strong classifiers [10], and the Random Forest result echoes the finding of Chen, Liaw, and Breiman that weighting is a weak lever for that family compared with balanced sampling [9].

4.3 Threshold Selection Works Everywhere, and ROC-AUC Explains Why

Validation-tuned thresholding on the default-trained model was the only strategy effective for every model family at every prevalence: balanced accuracy 0.706 to 0.748 at 2 percent and 0.737 to 0.768 at 10 percent, matching or exceeding the best training-time correction for each model (Gradient Boosting at 2 percent: threshold 0.706 vs. weighting 0.695, $p = 0.084$), and requiring no retraining. Combining weighting with threshold selection added essentially nothing over thresholding alone (Table 2).

The explanation is visible in the ranking metrics. Table 3 reports test ROC-AUC by training strategy: across models and prevalences, training-time corrections left ROC-AUC nearly unchanged, with differences of 0.00 to 0.02 (the largest single shift being undersampling raising Random Forest AUC from 0.775 to 0.798 at 2 percent). The models' learned rankings are largely intact even under severe imbalance; what fails is the fixed 0.5 cutoff applied to probabilities that concentrate near the low base rate. Correcting the operating point therefore recovers the achievable balanced accuracy without touching the model. PR-AUC values contextualize the difficulty: for default models, average precision was 0.075 to 0.113 at 2 percent prevalence (against a 0.02 chance baseline) and 0.78 to 0.82 at 50 percent, a reminder that PR-based metrics must be interpreted relative to prevalence [5].

Table 3. Test ROC-AUC by training strategy (mean over 10 seeds). Training-time corrections leave the learned ranking nearly unchanged; the failures and successes in Table 2 are operating-point effects.

Prevalence	Model	Default	Class weight	Undersample	SMOTE
2%	Logistic Regression	0.824	0.823	0.814	0.813
	Random Forest	0.775	0.776	0.798	0.778
	Gradient Boosting	0.788	0.794	0.781	0.783
10%	Logistic Regression	0.842	0.842	0.839	0.838
	Random Forest	0.809	0.809	0.815	0.805
	Gradient Boosting	0.833	0.831	0.817	0.821

4.4 The Selected Threshold Tracks the Prevalence

Figure 3 shows the median validation-selected threshold as a function of prevalence. For default-trained models the selected threshold approximately equals the prevalence itself: medians of 0.012 to 0.021 at 2 percent, 0.097 to 0.103 at 10 percent, and 0.46 to 0.50 at 50 percent, hugging the identity line across the full range. After effective class weighting, the selected threshold reverts toward 0.5, immediately for Logistic Regression (median 0.487 at 2 percent) and progressively for Gradient Boosting (0.279 at 2 percent, 0.481 by 10 percent), because weighting has already recalibrated the model toward a balanced prior. Random Forest is the diagnostic exception: its weighted-model threshold still tracks the prevalence (0.018 at 2 percent), confirming from a second direction that weighting never changed that ensemble's behavior. These observations align with the decision-theoretic account in which the optimal operating point is set by the effective class prior [7], and they yield a practical rule of thumb: for an uncorrected model on imbalanced data, a sensible starting cutoff is the prevalence, not 0.5.

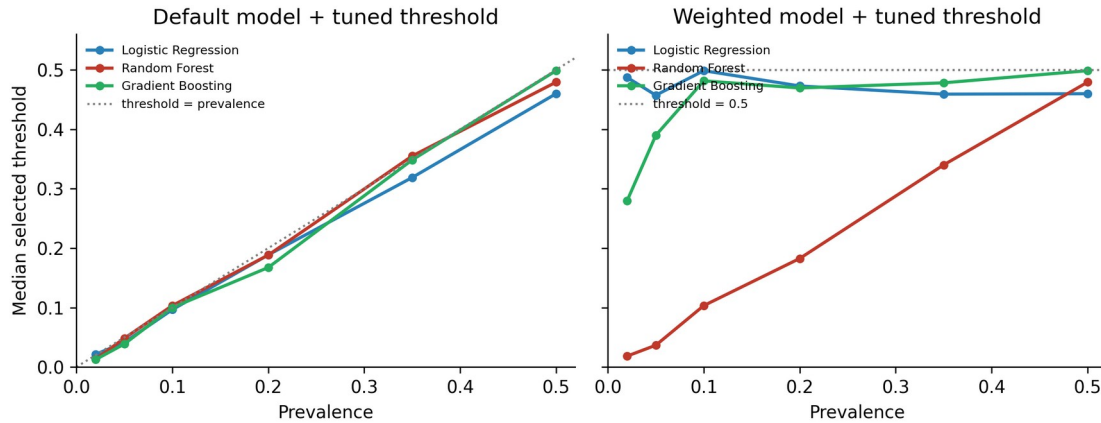


Figure 3. Median validation-selected Youden threshold versus prevalence. Left: default-trained models track the identity line (threshold approximately equals prevalence). Right: after class weighting, thresholds revert toward 0.5 for Logistic Regression and Gradient Boosting, while Random Forest's still tracks prevalence because weighting left it unchanged.

4.5 Replication on Two Additional Datasets and a Direct Test of the SMOTE Mechanism

Applying the identical protocol to a continuous-feature cardiovascular dataset and a binary-feature diabetes dataset reproduced every headline finding. On both datasets, default training failed at 2 percent prevalence for all three models (balanced accuracy 0.500), class weighting again failed completely for Random Forest (0.500 on both at 2 percent, unchanged from default), and validation-tuned thresholding again worked for every model at every prevalence tested. The model-family pattern is therefore not an artifact of the primary dataset.

The datasets also provide a direct test of the mechanism proposed for SMOTE's failure in Section 4.2, namely that interpolation harms tree models specifically when features are predominantly binary indicators. The three datasets bracket a range of continuous-feature share: the primary heart dataset has 4 continuous of 37 features, diabetes has 7 of 21, and cardiovascular has 5 of 11. As predicted, SMOTE's effectiveness for the tree ensembles rose with continuous-feature share. For Random Forest at 10 percent prevalence, SMOTE balanced accuracy was 0.582 on the primary (mostly binary) dataset and a comparable 0.588 on the similarly-structured diabetes dataset, but 0.622 on the continuous-dominant cardiovascular dataset, a clear recovery from that dataset's default of 0.536, whereas SMOTE barely moved the tree models on the two binary-dominant datasets. Figure 4 shows the pattern across all three datasets. This is empirical support for a specific, falsifiable mechanism rather than a dataset-specific observation: SMOTE should be avoided with tree ensembles on one-hot-encoded categorical data, but is more defensible when continuous features dominate.

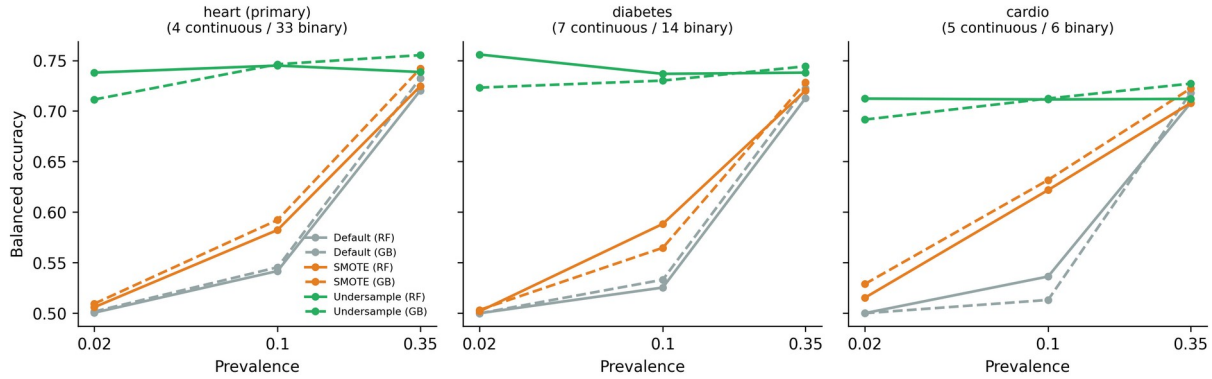


Figure 4. Balanced accuracy of SMOTE, undersampling, and default training for the two tree ensembles (RF solid, GB dashed) across three datasets ordered by increasing continuous-feature share (left to right). SMOTE recovers for the tree models only as continuous features come to dominate the feature space; undersampling works throughout.

4.6 Focal Loss

Focal loss, evaluated as an additional asymmetric training objective, behaved as a competent member of the class-weighting family rather than a superior alternative. For Logistic Regression on the primary dataset, focal loss matched balanced class weighting almost exactly (balanced accuracy 0.751 vs. 0.752 at 2 percent prevalence, paired $p = 1.0$) and tracked the weighting and threshold curves across all prevalence levels (Figure 5). For the boosted-tree implementation, focal LightGBM slightly exceeded class-weighted LightGBM at low prevalence (0.550 vs. 0.545 at 2 percent; 0.644 vs. 0.634 at 5 percent, paired $p = 0.010$), consistent with focal loss being designed for exactly the hard-minority regime, but at every prevalence it remained below validation-tuned thresholding of the same model (0.682 at 2 percent, 0.743 at 10 percent). Focal loss is thus a reasonable training-time correction where a custom objective is convenient, but it did not overturn the central finding that operating-point selection is the more reliable and lower-cost lever.

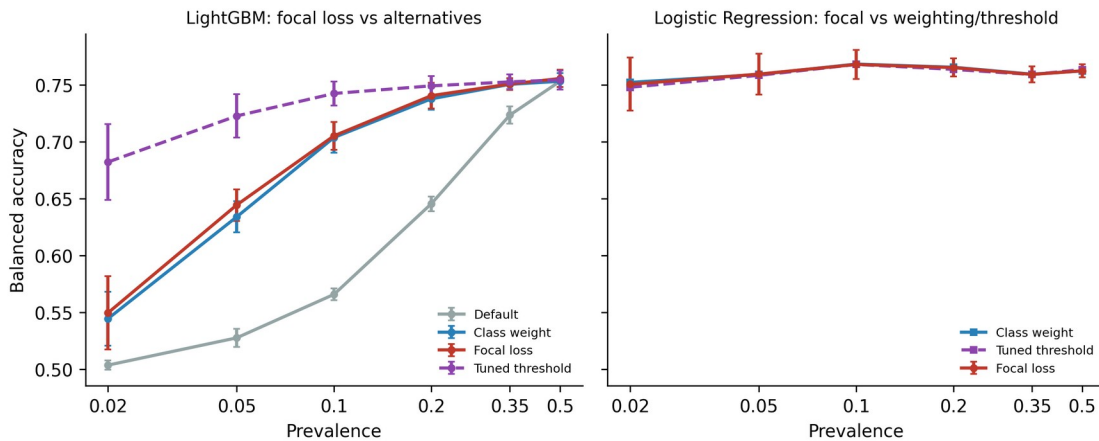


Figure 5. Focal loss compared with class weighting and validation-tuned thresholding. Left: LightGBM on the primary dataset. Right: Logistic Regression. Focal loss matches class weighting but does not exceed threshold tuning at any prevalence.

5. Discussion

The results support five general principles. First, at the prevalences studied, the class-imbalance problem is primarily an operating-point problem rather than a representation problem: training-time corrections barely changed ROC-AUC, and selecting a threshold on a validation split, with no retraining, recovered full balanced accuracy for every model at every prevalence. Second, training-time corrections are mechanism dependent. Corrections succeed where they shift the effective class prior seen by the learner's objective: class weighting enters the loss of Logistic Regression and Gradient Boosting and works for both, while it fails for the bagged majority-vote Random Forest, and undersampling, which changes the data prior directly, works for all three families. SMOTE's interpolation-based oversampling fails for tree models on predominantly one-hot feature spaces while matching weighting for the linear model, a mechanism confirmed across three datasets by the recovery of SMOTE as continuous features come to dominate. Third, validation-based threshold selection is the most robust single strategy available: uniformly effective, model agnostic, and free, and combining it with weighting, or replacing weighting with focal loss, adds nothing beyond it. Fourth, the selected threshold approximately equals the prevalence for default-trained models and reverts to 0.5 after effective reweighting, giving practitioners both a diagnostic (a tuned threshold far below 0.5 signals an uncorrected prior) and a starting heuristic. Fifth, corrections matter only below roughly 20 percent prevalence; by 35 percent every strategy converges, so effort spent on imbalance handling in mildly imbalanced settings is wasted.

These principles carry direct practical guidance for clinical screening models. Real screening populations sit at the low-prevalence end of this design, where the differences are largest, so the choice of correction genuinely matters there. For tree ensembles, the two most commonly recommended fixes, class weighting and SMOTE, were the two that failed, while the simplest alternatives, random undersampling and threshold tuning, succeeded. The threshold-only approach has an additional advantage highlighted by van den Goorbergh et al. [8]: because it leaves the trained model untouched, it preserves whatever calibration the model has, addressing imbalance at decision time instead of distorting the risk estimates themselves. Finally, the corrected linear model matched or slightly exceeded both corrected ensembles throughout (balanced accuracy ceilings of 0.748 to 0.768 versus 0.706 to 0.757), consistent with reports that well-specified logistic regression remains competitive on tabular clinical data [8].

The study also illustrates a reporting practice worth adopting: single-prevalence, single-strategy evaluations can badly misrepresent a model family. Random Forest with class weighting looks like a reasonable published configuration, yet here it was never better, and once significantly worse, than doing nothing. Mapping performance across prevalence levels, with matched sample sizes and repeated draws, is what made the failure modes visible.

6. Limitations

Several limitations bound these conclusions. The main sweep used one dataset and one feature space, and although two further datasets were added for replication and to test the SMOTE mechanism, the replication covered three prevalence levels with five seeds rather than the full six-by-ten design, and all three datasets remain within health and clinical survey data. One dataset size ($N = 20,000$) and one threshold objective (Youden's J) were used throughout; calibration-based operating-point methods, cost-weighted threshold objectives reflecting asymmetric clinical consequences, and per-tree balanced Random Forest variants [9] were not evaluated. Focal loss was tested for two model types on the primary dataset only. Prevalence was manipulated by subsampling, which changes the class mix but not the within-class covariate distributions, so real

population shifts between prevalence regimes are not modeled. Statistical tests were limited to a small set of pre-specified contrasts with modest replicate counts and no multiple-comparison correction. Finally, the underlying records are self-reported or examination-based survey responses, so the models capture associations rather than diagnostic relationships, and absolute performance levels should not be read as clinical benchmarks.

7. Conclusion

A controlled factorial study of class-imbalance handling, spanning six prevalence levels, three model families, six strategies, and ten replicates per cell on 319,795 clinical survey records, shows that the standard corrections are far from interchangeable. Default training fails at low prevalence for every family and needs no correction at all above roughly 35 percent prevalence. Class weighting fails completely for Random Forest while working for loss-based learners; SMOTE fails for tree ensembles on predominantly categorical features while matching weighting for the linear model; random undersampling works everywhere; and validation-based threshold selection works everywhere at zero retraining cost, because the corrections were never improving the learned ranking in the first place, only the operating point. The selected threshold approximately equals the prevalence, tying the empirical results to the decision-theoretic account and yielding a usable rule of thumb. Replication on two further datasets reproduced these patterns and confirmed, as a falsifiable prediction, that SMOTE's failure for tree ensembles is specific to binary-dominant feature spaces, and a focal-loss experiment placed that objective on par with class weighting but below threshold tuning. Future work will extend the replication to the full design and to non-clinical domains, add calibration-aware operating-point selection and balanced ensemble variants, and examine cost-weighted threshold objectives that reflect asymmetric clinical consequences.

Acknowledgments

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Data Availability and Ethics

This study used de-identified, publicly available survey data [3], [11]. No human subjects were directly involved, and no personally identifiable information was used. The complete experiment code, the per-run results for all 1,080 evaluations, and the analysis notebook are available from the author.

Conflict of Interest

The author declares no conflict of interest.

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